

Weekly Homework 02: Data-Driven Programs

Course: Python Programming

Textbook: *Murach's Python Programming*, 2nd edition

Related class activity: Activity 2, Trivia Night

Suggested time: 2-3 hours

Total: 100 points

Purpose

In class, you built a trivia game that stores questions as structured data and processes them with functions and loops. This homework gives you individual practice with lists, dictionaries, functions, and input validation using the textbook's exercises.

Read and Complete

Use the exercise files that accompany your textbook, if provided. Follow the book's instructions for opening and running each program. The page numbers below are the printed page numbers in the 2nd edition.

A. Required: Dictionaries — Exercise 12-2, pages 353-354

Complete steps 3-7 of "**Enhance the Movie List 2D program.**" Convert each movie record to a dictionary while keeping the collection of movies as a list. Update the list, add, and delete behavior so the program still works.

This is the main assignment because it practices the same data shape used by the trivia question bank: a list containing dictionaries with named fields.

B. Required: Lists and Functions — Exercise 6-1, pages 199-200

Complete steps 3-6 of "**Use a list for the Test Scores program.**" Store the scores in a list, pass that list between functions, and calculate the total, count, and average. Step 7's additional statistics are optional enrichment.

C. Required: Exceptions — Exercise 8-1, page 251

Complete steps 3-4 of "**Add exception handling to the Future Value program.**" Test invalid numeric entries, add exception handling, and test again. Pay attention to the difference between invalid text that cannot be converted to a number and a number that is outside the allowed range.

What to Submit

Submit one clearly named folder or ZIP containing:

- The completed Python files for all three exercises.
- A README.md with your name, the exercise numbers, how to run each program, and a short note describing one problem you found and how you fixed it.
- A short AI-use note: name the tool(s) used, describe what you asked, and identify one suggestion you accepted, changed, or rejected. If you did not use AI, say so.

Do not submit the textbook PDF, a classmate's code, passwords, API keys, or private information. Keep the original textbook exercise starter files intact by working on copies.

Using AI for Homework

AI is allowed as a learning and review tool. You are responsible for the submitted work and must be able to explain it.

Good requests include:

- "Explain what this traceback is telling me. Give me one hint, not a complete rewrite."
- "Review my function call and explain why the value is not being returned as expected."
- "What tests should I try for this input validation? Do not write the solution."

Do not ask AI to complete the textbook exercises for you or submit code you cannot explain. Verify any generated advice by running your program and checking the result. Disclose meaningful AI assistance in your README.

Rubric (100 points)

Area	Points	What earns credit
Exercise 12-2: list of dictionaries	30	Each record uses the required dictionary fields; list, add, and delete operations continue to work.
Exercise 6-1: list and function flow	25	Scores are collected in a list, passed to the processing function, and used to calculate correct total, count, and average.
Exercise 8-1: exception handling	20	Invalid number text is handled without an unexpected crash; valid and invalid entries are tested; range validation remains correct.
Testing and correctness	10	Programs run; evidence or notes cover normal and relevant edge cases; reported output is consistent with the code.

Area	Points	What earns credit
README and organization	10	Files are clearly named and organized; README includes run instructions and one debugging reflection.
AI-use disclosure and explanation	5	AI use is honestly disclosed, or non-use is stated; student can describe the submitted work.

Partial credit is available when a program is incomplete but the work demonstrates a correct attempt. A copied or unexplained solution may receive no credit for the affected criteria, consistent with course policy.

Before You Submit

- Run each program from a fresh start.
- Test both an ordinary case and at least one invalid or boundary case where relevant.
- Make sure your README tells me how to run the files.
- Confirm you submitted your own work and included your AI-use note.